

MOHAMMAD ARIAIMEHR

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PROFESSIONAL SUMMARY

Research-oriented software engineer with experience in Human Activity Recognition, Deep Learning, and Federated Learning. Author of a Q1 journal publication on transformer-based HAR. Interested in multimodal AI, computer vision, wearable sensing, and AI-driven human motion analysis using video and sensor data.

EDUCATION

Azad University, Tehran Central Branch Tehran, Iran
Master of Science in Software Engineering January 2021 - April 2024

- Thesis: "Human Activity Recognition with Applied Positional Encoding on Attention Matrix on Each Attention Head." Score: 18.50/20 (4/4)
- CGPA: 15.6/20 (Equivalent GPA: 3.31/4)

Azad University, Marand Branch Marand, Iran
Bachelor of Science in Electronic Technology Engineering September 2012 - July 2014

- Completed entrepreneurship project modeling low-power comparisons with a supply voltage of 0.5 V and a frequency of 2 MHz. Score: 18/20 (4/4)
- CGPA: 17.09/20 (Equivalent GPA: 3.55/4)

Azad University, Soufian Branch Soufian, Iran
Associate Degree in General Electronics September 2010 - July 2012

- Designed and simulated an analog-to-digital converter based on a low-voltage comparator for the entrepreneurship project. Score: 20/20 (4/4)
- CGPA: 17.61/20 (Equivalent GPA: 3.64/4)

RESEARCH INTERESTS

Core Interests: Human Activity Recognition, Computer Vision, Machine Learning, Multimodal AI, Wearable Computing, Human Motion Analysis

PUBLICATIONS

Ariaimehr, M. and Ravanmehr, R. (2025). Human activity recognition using transformer-based positional encoding in a federated learning framework. *Multimedia Tools and Applications*, 84, 50199–50248. <https://doi.org/10.1007/s11042-025-21086-5>

Research Keywords: Human Activity Recognition, Transformer Networks, Wearable Sensors, Time-Series Analysis, Deep Learning, Federated Learning

PEER REVIEW ACTIVITY

Reviewer, Expert Systems with Applications (Elsevier) September 2026

- Awarded Certificate of Reviewing in recognition of the review contributed to the journal.

RESEARCH EXPERIENCES

Azad University, Tehran Central Branch Tehran, Iran
Human Activity Recognition Using Transformer Model 2022 - 2024

- Developed a transformer-based framework for wearable sensor Human Activity Recognition.
- Processed multivariate wearable IMU time-series collected from public HAR datasets (PAMAP2, MotionSense, and Opportunity).

- Designed deep learning models for human movement analysis and evaluated performance using Accuracy, Precision, Recall and F1-score.
- Integrated Federated Learning for privacy-preserving distributed model training.
- Trained and evaluated transformer-based models on real-world human movement data.

TECHNICAL SKILLS

- **Programming:** Python
- **AI Frameworks & Data Processing:** TensorFlow, Keras, NumPy, Pandas
- **Machine Learning:** Deep Learning, Federated Learning, Transformer Models, Time-Series Analysis
- **DevOps & OS:** Linux, Git

WORK EXPERIENCES

Behpardaz Hamrah Samaneh Aval (Behsa) September 2020 - Present
System Engineer Tehran, Iran

- Maintained and managed Linux-based servers for Mobile Communications Company of Iran, reducing server downtime and enhancing security measures against potential cyber threats.
- Established and maintained Linux servers for Rightel and managed their OCS system.
- Automated Linux administration tasks using Bash and Python scripting to improve operational efficiency.

Red Crescent Society April 2014 - June 2014
Intern Tehran, Iran

- Collaborated on developing a rescue robot designed for search operations under rubble.
- Integrated cameras, thermal sensors, and gas detectors to assist the robot in identifying survivors or hazardous conditions.

Telecommunications Company May 2012 - July 2012
Intern Tehran, Iran

- Worked in the Main Distribution Frame (MDF), managing cables, installing equipment, and troubleshooting issues.

ACADEMIC PROJECTS

Lightweight Operating System for IoT | Advanced Operating Systems Course Sept - Dec 2021

- Contributed to developing a lightweight operating system tailored for IoT devices. Project grade: 17/20 (4/4).

Microservices Architecture Design | Software Architecture Course Feb - Jun 2022

- Designed a microservices architecture for a cloud-based application scaling effectively with user demand. Project grade: 17/20 (4/4).

CERTIFICATES

Online Courses: Deep Learning for Business, Coursera / Yonsei University (2024); Python for Data Science, AI, and Development, Coursera / IBM (2023); Neural Networks and Deep Learning, Coursera / DeepLearning.AI (2022); Linux Server Management and Security, Coursera / Univ. of Colorado (2022); DevOps and Build Automation with Python, Coursera / LearnQuest (2022)

In-Person Courses (CanDo Institute): DevOps for Intermediates (2023), DevOps for Beginners (2022), CCNA (2022), Security+ (2022), LPIC2 (2022), Python Programming (2021), LPIC1 (2020)

LANGUAGES

- **English:** IELTS Academic - Overall Band 7.0 (CEFR Level C1). Reading: 7.5 | Writing: 7.0 | Listening: 7.0 | Speaking: 7.0 (Awarded: 30 Sept 2025)
- **German:** A1 level (Ava-e-Chista Institute)
- **Italian:** B2 level (University for Foreigners of Siena)